Assignment-3

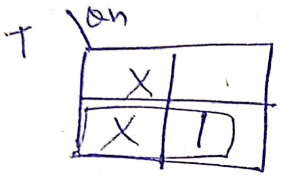
(1) Show how to convert JK flip flop into
a) T-flip flop (b) D-flip flop.

T	Q_n	Q_{n+1}	J	K
0	0	0	0	X
1	0	1	1	X
1	1	0	X	1
0	1	1	X	0

$J = \Sigma m(2) + \phi(3,1)$

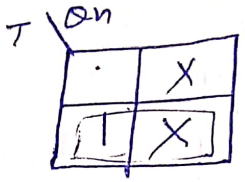
$K = \Sigma m(3) + \phi(0,2)$

To find K:-

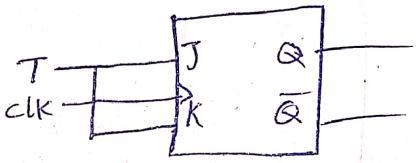


$K = T$

To find J:-



$J = T$



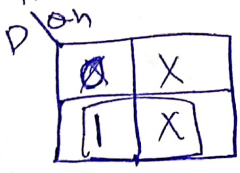
b) D-flip flop

D	Q_n	Q_{n+1}	J	K
0	0	0	0	X
1	0	1	1	X
0	1	0	X	1
1	1	1	X	0

$J = \Sigma m(2) + \phi(1,3)$

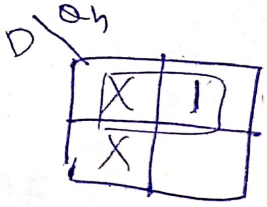
$K = \Sigma m(1) + \phi(0,2)$

To find J:-

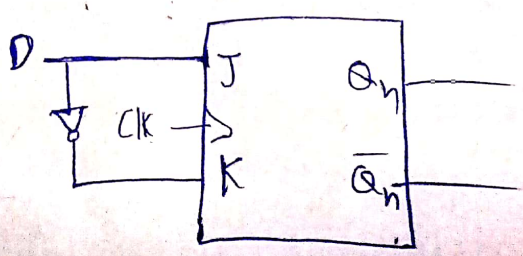


$J = D$

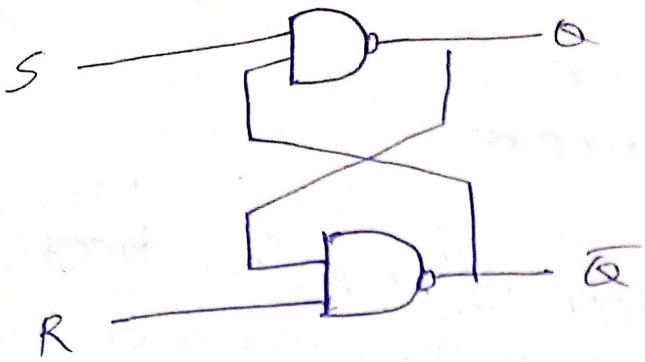
To find K:-



$K = \overline{D}$

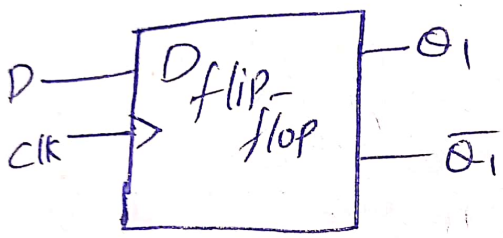


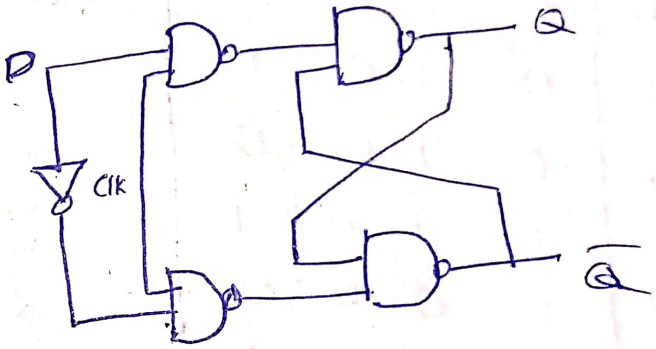
(2) Implement SR Latch with control i/p using 2-i/p NAND gates



S	R	Q_n
0	0	Undefind
0	1	1
1	0	0
1	1	No change

(3) Draw the circuit for D flip flop with Synchronous Reset?





(4) which flip flop can be used as delay switch and as toggle switch?

Sol: The JK flip flop's flexibility in handling different i/p combinations allows it to be configured to perform both delay and toggle function.

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(6) Design mod 10 counter using Asyn and synchronous counter?

Sol:- Asynchronous:-

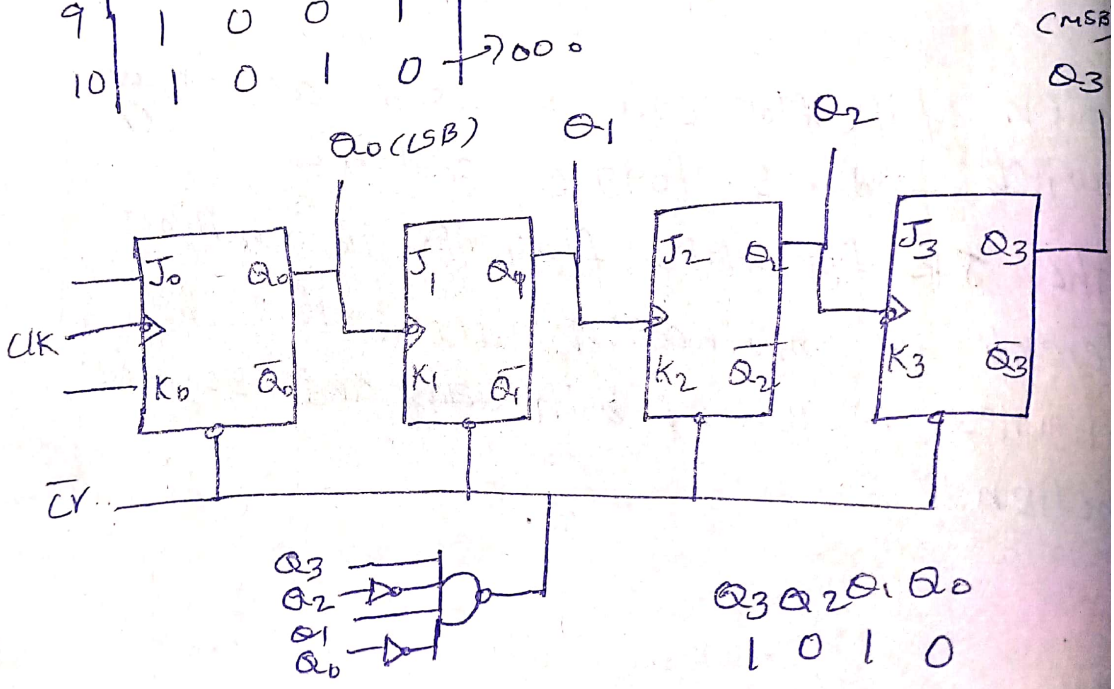
$N = 10$

$2^n \geq N \Rightarrow 2^n \geq 10 \Rightarrow n = 4$

4 f/f are required.

CLK	Q_3	Q_2	Q_1	Q_0
0	0	0	0	0
1	0	0	0	1
2	0	0	1	0
3	0	0	1	1
4	0	1	0	0
5	0	1	0	1
6	0	1	1	0
7	0	1	1	1
8	1	0	0	0
9	1	0	0	1
10	1	0	1	0 → 0000

after 10th CLK we need $Q_3Q_2Q_1Q_0 = 0000$ i.e expeted state 1010 has to become 0000



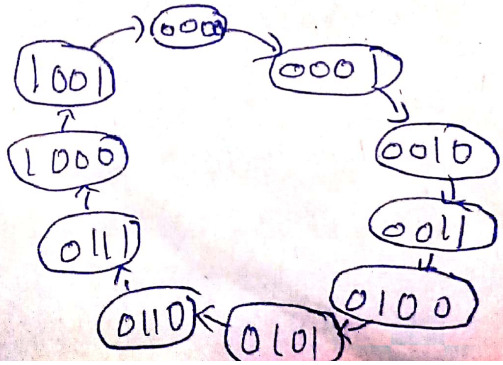
$Q_3Q_2Q_1Q_0$

1 0 1 0

* Synchronous counter:-

$N = 10$

$2^n \geq N \Rightarrow 2^n \geq 10 \Rightarrow n = 4$



PS				NS											
Q_A	Q_B	Q_C	Q_D	Q_{A+1}	Q_{B+1}	Q_{C+1}	Q_{D+1}	J_A	K_A	J_B	K_B	J_C	K_C	J_D	K_D
0	0	0	0	0	0	0	1	0	X	0	X	0	X	1	X
0	0	0	1	0	0	1	0	0	X	0	X	1	X	X	1
0	0	1	0	0	0	1	1	0	X	0	X	X	0	1	X
0	0	1	1	0	1	0	0	0	X	1	X	X	1	X	1
0	1	0	0	0	1	0	1	0	X	X	0	X	X	1	X
0	1	0	1	0	1	1	0	0	X	X	0	0	X	X	1
0	1	1	0	0	1	1	1	0	X	X	0	1	X	X	1
0	1	1	1	1	0	0	0	1	X	X	1	X	0	X	X
1	0	0	0	1	0	0	1	X	0	0	X	X	1	1	1
1	0	0	1	0	0	0	0	X	1	0	X	0	X	X	
1	0	1	0	X	X	X	X	X	X	X	X	X	X	X	X
1	0	1	1	X	X	X	X	X	X	X	X	X	X	X	X
1	1	0	0	X	X	X	X	X	X	X	X	X	X	X	X
1	1	0	1	X	X	X	X	X	X	X	X	X	X	X	X
1	1	1	0	X	X	X	X	X	X	X	X	X	X	X	X
1	1	1	1	X	X	X	X	X	X	X	X	X	X	X	X

$J_A = \Sigma m(7) + \phi(8,9,10,11,12,13,14,15)$

$K_A = \Sigma m(9) + d(0,1,2,3,4,5,6,7,10,11,12,13,14,15)$

$J_B = \Sigma m(3) + d(4,5,6,7,10,11,12,13,14,15)$

$K_B = \Sigma m(7) + d(0,1,2,3,8,9,10,11,12,13,14,15)$

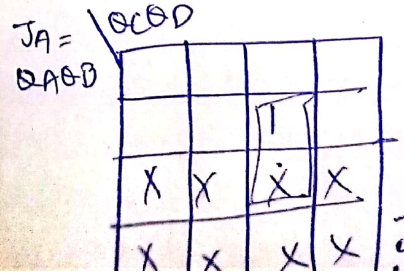
$J_C = \Sigma m(1,5) + d(2,3,6,7,10,11,12,13,14,15)$

$K_C = \Sigma m(3,7) + d(0,1,4,5,8,9,10,11,12,13,14,15)$

$J_D = \Sigma m(0,2,4,6,8) + d(1,3,5,7,9,10,11,12,13,14,15)$

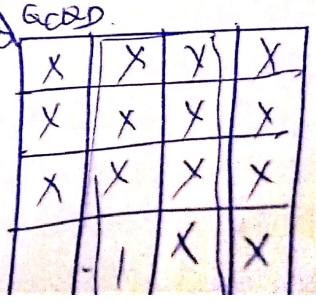
$K_D = \Sigma m(1,3,5,7,9) + d(0,2,4,6,8,10,11,12,13,14,15)$

$J_A =$



$J_A = Q_B Q_C Q_D$

J_B



J_B

$Q_A Q_B$ \ $Q_C Q_D$	00	01	11	10
00			1	
01	X	X	X	X
10	X	X	X	X
11			X	X

K_B

$Q_A Q_B$ \ $Q_C Q_D$	00	01	11	10
00	X	X	X	X
01			1	
10	X	X	X	X
11	X	X	X	X

$J_B = Q_C Q_D$

$K_B = Q_C Q_D$

J_C

$Q_A Q_B$ \ $Q_C Q_D$	00	01	11	10
00		1	X	X
01		1	X	X
11	X	X	X	X
10			X	X

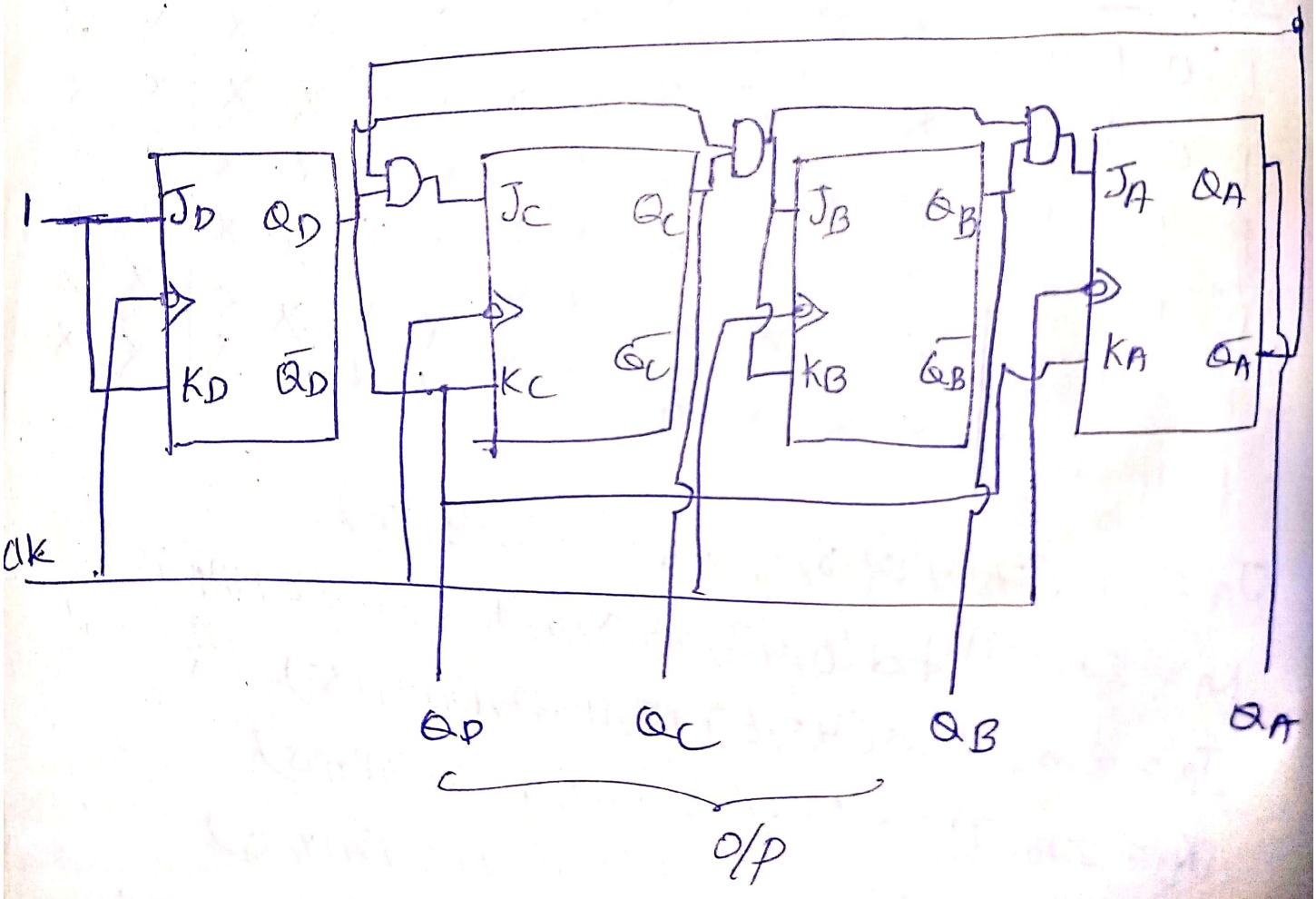
K_C

$Q_A Q_B$ \ $Q_C Q_D$	00	01	11	10
00	X	X	1	
01	X	X	1	
11	X	X	X	X
10	X	X	X	X

$J_C = \overline{Q_A} Q_B Q_D$

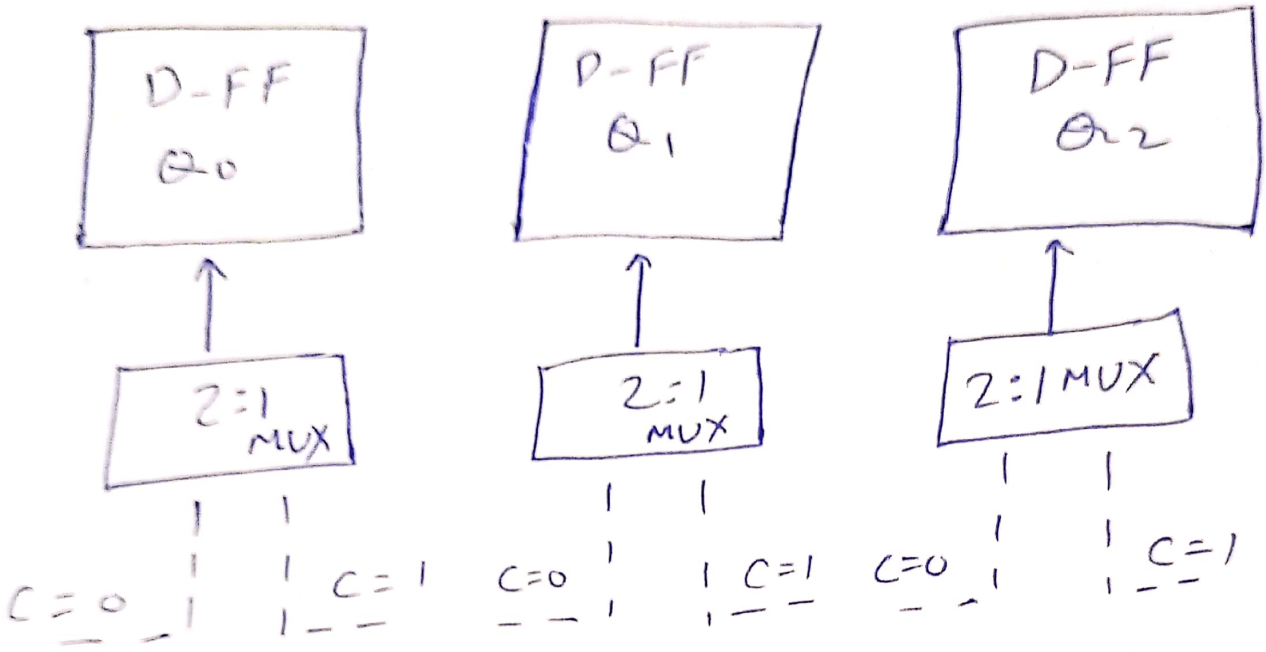
$K_C = Q_D$

$J_D = K_D = 1$



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(5) Design a 3 bit shift register using 2:1 MUX and DFF which shifts right if Control i/p C=0 & shift left if Ctrl i/p C=1



=) if C = 0 it shift right
each mux select i/p from right

=) if C=1
each mux select i/p from left

=) MUX i/p 10 and 11 and C is the selection line

10 → right shift
11 → Left shift
C → it may shift right (or) left based on the i/p.